

300,000 Streets of Melbourne

School Streets Safety — Quick Revision Sheet

The pitch

A reproducible Python pipeline that **scores**, **maps**, and **predicts** the safety of school walking routes using open government data and the **Healthy Streets** framework — then surfaces costed fixes for councils.

Numbers to remember

192%	crash increase in Darebin LGA, 2021 → 2024 (25 → 73)
0.4	Preston HS HS2 score — no safe crossing
r=0.84	Pearson correlation: SEIFA decile vs HS score
100%	William Ruthven SC crashes during school hours
7,948	Victorian crash records analysed (2021–2025)
26	open-data features fed into ML model
MAE	Ridge regression mean absolute error (LOO-CV, n=3)
2.88	
30	pre-computed scenario results (10 templates × 3 schools)
\$80k	cheapest fix for Preston HS (signalised crossing)
200+	Melbourne secondary schools scalable to with no code changes

10 Healthy Streets indicators (threshold: 6.0 / 10)

Code	Indicator	Source	Worst score
HS1	Footpath	Field	Reservoir 4.2 [MOD]
HS2	Easy to cross	Field	Preston 0.4 [MAJOR]
HS3	Shade	OSM	W.Ruthven 3.0 [MOD]
HS4	Rest places	OSM	W.Ruthven 3.3 [MOD]
HS5	Not too noisy	Field	Reservoir 5.0
HS6	Active travel	OSM+fld	W.Ruthven 4.2
HS7	Feel safe	CSA	All ≥ 7.5
HS8	Things to do	OSM	W.Ruthven 4.3
HS9	Feel relaxed	Field	Reservoir 5.2
HS10	Clean air	EPA	All ≥ 9.0

Severity rules: **Major** if HS2 < 4 OR HS1 < 4 OR HS5 < 3 • **Moderate** if 2+ indicators < 6.0

3 schools at a glance

School	Overall	Severity	Key issue
Preston HS	7.2	MAJOR	HS2=0.4 no crossing
Reservoir HS	6.1	MOD	HS1=4.2 broken footpath
W. Ruthven SC	6.7	MOD	HS3=3.0 no shade

*The paradox: Preston has the **highest** overall score but the **only Major** severity. One failing indicator overrides nine green ones.*

Pipeline — 10 steps, 1 command

python run_all.py

- Ingest** — crash (VicRoads), streets (OSM), air (EPA), crime (CSA), SEIFA, Census
- Feature engineering** — 26 features per school gate; filter sentinels 777/888/999
- HS scoring** — src/scoring/hs1.py ... hs10.py; each scorer

is independent

- Severity classify** — rule-based thresholds in **severity.py**
- Recommendations** — rule engine: indicator gap → intervention + cost + timeline
- ML training** — Ridge regression, LOO-CV, 26 features, predict HS1–HS10
- Scenario engine** — 10 templates × 3 schools = 30 delta results
- Equity & crash analysis** — SEIFA join, 400 m crash buffers, hourly breakdown
- Visualisation** — 9 charts (Matplotlib); GeoPackage + QGIS project export
- Dashboard export** — JSON → static GitHub Pages (Leaflet + Chart.js)

Machine learning — what and why

- **Algorithm:** Ridge regression (handles collinear OSM features)
- **Features:** 26 (20 spatial via OSM, 2 environmental EPA/CSA, 4 crash stats)
- **Evaluation:** Leave-One-Out CV — only valid method with $n=3$ schools
- **Mean MAE:** 2.88 across all 10 indicators
- **Can automate:** HS7 (crime data) and HS10 (EPA air quality) — no surveyor needed
- **Cannot automate:** HS2 — crossing *presence* $\neq$ crossing *quality*

Implication: councils can triage all 200+ Melbourne schools on HS7 and HS10 at zero survey cost today.

Equity finding

School	SEIFA	HS	Walking dependency
Reservoir HS	3.7	6.1	High — lower income, less cars
W.Ruthven SC	3.7	6.7	High — similar catchment
Preston HS	5.5	7.2	Lower — more car ownership

$r = 0.84$: lower decile (more disadvantage) $\rightarrow$ worse safety score. Lower income = less car access = more walking = more exposure to dangerous streets.

Scenario engine — top 3 fixes

School	Intervention	Cost	Impact
Preston HS	Signalised crossing	\$80–200k	HS2: 0.4 $\rightarrow$ 3.5; MA-JOR $\rightarrow$ MOD
Reservoir HS	Footpath repair	\$100–300k	HS1: 4.2 $\rightarrow$ 7.0
W.Ruthven SC	40 km/h school zone	\$5–20k	HS9: instant gain

Codebase map

src/scoring/	hs1.py ... hs10.py + severity.py
src/features/	engineer.py (26 features)
src/ml/	train / predict / evaluate
src/scenarios/	10 intervention delta templates
src/recommendations/	rule-to-fix mapping + cost metadata
src/data/	download + cache handlers
src/visualisation/	9 chart builders
docs/	GitHub Pages dashboard
outputs/	CSV, PNG, GPKG, HTML, PPTX

Tech stack (one line)

Python 3.12 • pandas / geopandas / OSMnx • scikit-learn • Matplotlib • Leaflet.js • Chart.js • GitHub Pages • L^AT_EX

Likely Q&A

- **Why Ridge?** Collinear OSM features; Ridge handles this; interpretable weights.
- **Why LOO-CV?** Only 3 schools — any train/test split would give 0 test samples.
- **Why not random forest?** $N=3$ gives insufficient variance for ensemble methods.
- **How does it scale?** Same `run_all.py`; swap school list in `config.py`; no code changes.
- **Is the $r=0.84$ significant?** With $n=3$ it's directional not inferential — acknowledged in analysis.
- **Why not HS2 automated?** OSM tags crossing presence; it cannot tag signal timing, visibility, sightlines, or informal crossings.
- **What does SEIFA measure?** Relative socio-economic disadvantage (ABS 2021, SA1 level, pop-weighted IRSD).
- **Delta method for scenarios?** Anchors to real observed scores; model estimates the *change*, not the absolute value — avoids compounding prediction error.

Dashboard: s4111078-sidd.github.io/300-000-School-Streets

Run everything: `python run_all.py`